

i)

Given

$$r = .001 \text{ m} \quad L = .05 \text{ m} \quad k = 200 \text{ W/m K}$$

Adiabatic tip

Schematic below

Find

a) $T_{L/2} = ?$, $T_L = ?$ if $T_b = 125^\circ\text{C}$

b) ξ_f , η_f by Table 3.4

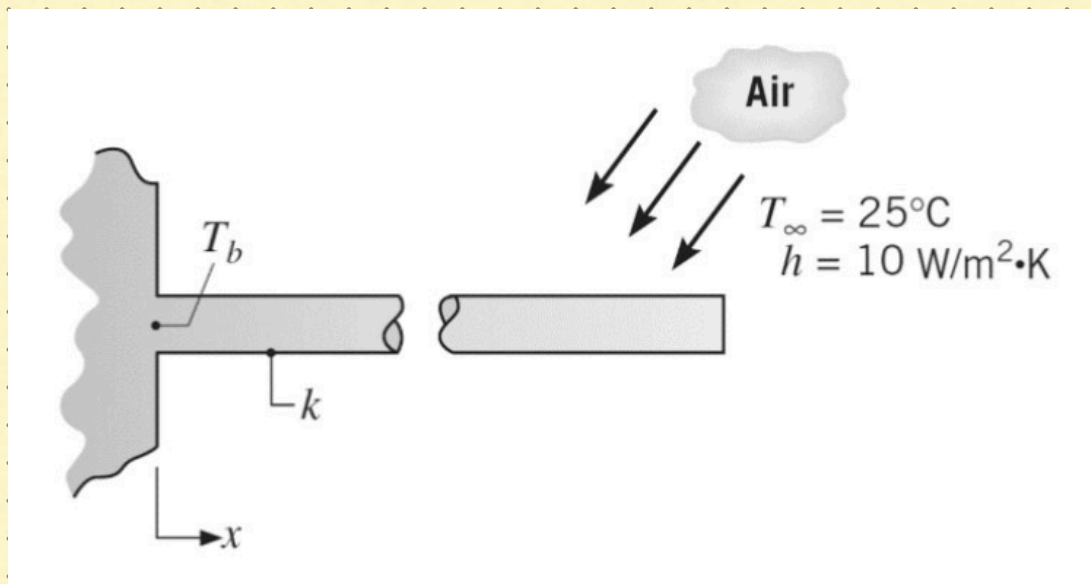
c) η_f by Table 3.5. Compare to part (b)

d) $R_f = ?$

e) $\eta_o = ?$, R_o

f) Resistance network, $T_h = ?$. Compare to no heat sink.

Schematic



Assumptions

ss, 1D heat flow, uniform cross section.

Analysis

a) From 3.4:

$$\frac{\theta}{\theta_b} = \frac{\cosh m(L-x)}{\cosh mL}$$

$$m = \sqrt{\frac{(10 \text{ W/m}^2\text{K})(.006 \text{ m})}{(200 \frac{\text{W}}{\text{mK}})(3.14 \text{E-}6 \text{ m}^2)}}$$

$$m = 9.77 \text{ 1/m}$$

$$\theta_{L/2} = (398 \text{ K} - 298) \frac{\cosh(9.77(.05 - .025))}{\cosh(9.77 \cdot .05)}$$

$$\theta_{L/2} = 91.82 \text{ K} = T_{L/2} - T_{\infty} = T_{L/2} - 298 \text{ K}$$

$$T_{L/2} = 389.82 \text{ K} = 116.82^{\circ}\text{C}$$

$$\theta_L = (398 \text{ K} - 298) \frac{\cosh(9.77(0))}{\cosh(9.77 \cdot .05)} = 89.15 \text{ K}$$

$$\theta_L = T_L - T_{\infty} = 89.15 \text{ K}$$

$$T_L = 387.15 \text{ K} = 114.15^{\circ}\text{C}$$

$$\theta = T - T_{\infty}$$

$$m = \sqrt{\frac{hP}{kA_c}}$$

$$P = 2\pi r = 2\pi(.001 \text{ m})$$

$$P = .006 \text{ m}$$

$$A_c = \pi r^2 = \pi(.001 \text{ m})^2$$

$$A_c = 3.14 \text{E-}6 \text{ m}^2$$

$$b) \quad \epsilon_f = \frac{q_f}{q_{inf}}$$

$$q_f = M \tanh(mL) \quad M = \sqrt{hPkA_c} \theta_b$$

$$q_{inf} = hA_c \theta_b$$

$$\epsilon_f = \frac{\sqrt{hPkA_c} \theta_b \tanh(mL)}{hA_c \theta_b}$$

$$\epsilon_f = \frac{\sqrt{(0.006 \text{ m})(200 \text{ W/mK})} \tanh(9.77 \cdot .05)}{\sqrt{(10 \text{ W/m}^2\text{K})(3.14 \text{ E-}6 \text{ m}^2)}}$$

$$\epsilon_f = 88.56$$

$$\eta_f = \frac{q_f}{q_{max}}$$

$$q_{max} = hA_f \theta_b$$

$$A_f = PL = (.006 \text{ m})(.05 \text{ m}) = 3 \text{ E-}4 \text{ m}^2$$

$$\eta_f = \frac{\sqrt{hPkA_c} \theta_b \tanh(mL)}{hA_f \theta_b}$$

$$\eta_f = \frac{\sqrt{(0.006 \text{ m})(200 \frac{\text{W}}{\text{mK}})(3.14 \text{ E-}6 \text{ m}^2)} \tanh(9.77 \cdot .05)}{\sqrt{(10 \text{ W/m}^2\text{K}) (3 \text{ E-}4 \text{ m}^2)}}$$

$$\eta_f = .927 = 92.7 \%$$

c) From Table 3.5 for pin fin:

$$\eta_f = \frac{\tanh(mL_c)}{mL_c} = \frac{\tanh(10 \cdot .0505)}{10 \cdot .0505}$$

$$m = \sqrt{\frac{4h}{KD}} = 10$$

$$L_c = .05 + \frac{.002}{4}$$

$$\eta_f = .923 = 92.3 \%$$

The expression for pin fin efficiency given in Table 3.5 is a great approximation for η_f .

d)

$$R_f = \frac{1}{\eta_f h A_f} = \frac{1}{(.927)(10 \frac{W}{m^2K})(.006m \cdot .05m)}$$

$$R_f = 359.58 \text{ K/W}$$

e)

$$\eta_o = 1 - \frac{NA_f}{A_{tot}} (1 - \eta_f)$$

$$\eta_o = 1 - \frac{(25)(.006 \cdot .05)}{.0083 \text{ m}^2} (1 - .927)$$

$$\eta_o = .934 = 93.4\%$$

$$R_o = \frac{1}{\eta_o h A_{tot}}$$

$$R_o = \frac{1}{(.934)(10 \frac{W}{m^2K})(.0083 \text{ m}^2)} = 12.87 \frac{K}{W} = R_o$$

$$\begin{aligned} A_{uf} &= s^2 - NA_c \\ &= (.03 \text{ m})^2 - (25)(\pi(.001)^2) \\ &= 8.21 \text{ E-4 m}^2 \end{aligned}$$

$$A_{tot} = NA_f + A_{uf}$$

$$A_{tot} = (25)(.006 \cdot .05) + 8.21 \text{ E-4 m}^2$$

$$A_{tot} = .0083 \text{ m}^2$$

$$q_o = 25q_f + q_{uf}$$

$$q_f = \frac{\Delta T}{R_f} = \frac{T_b - T_{\infty}}{R_f} = \frac{100 \text{ K}}{359.58 \text{ K/W}}$$

$$q_f = .278 \text{ W}$$

$$q_o = \frac{\Delta T}{R_o} = \frac{T_b - T_{\infty}}{R_o} = \frac{100 \text{ K}}{12.87 \text{ K/W}}$$

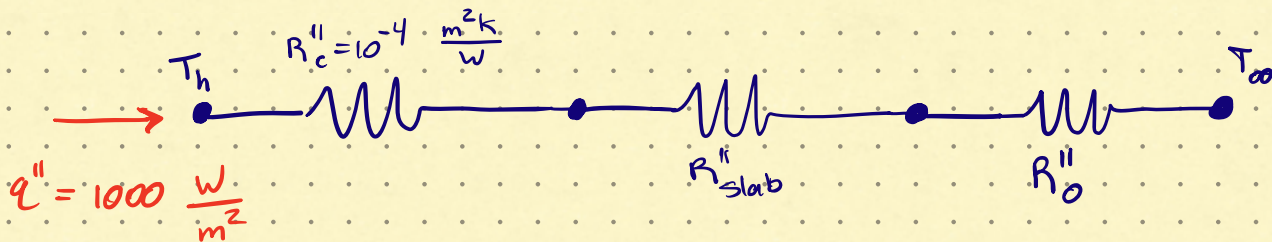
$$q_o = 7.77 \text{ W}$$

$$7.77 \text{ W} = 25(.278 \text{ W}) + q_{uf}$$

$$\Rightarrow q_{uf} = .817 \text{ W} \quad \text{Heat thru unfinned surface}$$

$$q_{f, \text{total}} = 25(.278 \text{ W}) = 6.95 \text{ W} \quad \text{Heat thru fins}$$

f)



$$R_{\text{slab}}'' = \frac{L}{k} = \frac{.02 \text{ m}}{10 \text{ W/mK}} = .002 \frac{\text{m}^2 \text{K}}{\text{W}}$$

$$R_o = 12.87 \frac{\text{K}}{\text{W}}$$

$$R_o'' = R_o s^2 = (12.87 \frac{\text{K}}{\text{W}})(.03 \text{ m})^2 = .012 \frac{\text{m}^2 \text{K}}{\text{W}}$$

$$R_{\text{eq}}'' = R_c'' + R_{\text{slab}}'' + R_o'' = .0141 \frac{\text{m}^2 \text{K}}{\text{W}}$$

$$\Delta T = q'' R_{eq}'' = \left(1000 \frac{\text{W}}{\text{m}^2}\right) \left(0.0141 \frac{\text{m}^2 \text{K}}{\text{W}}\right)$$

$$T_h - T_\infty = 14.1 \text{ K}$$

$$T_h = 312.1 \text{ K} \quad \text{with heat sink}$$

$$T_h = q'' (R_c'' + R_{slab}'' + R_{conv}'') + T_\infty = \left(1000 \frac{\text{W}}{\text{m}^2}\right) \left(0.0021 \frac{\text{m}^2 \text{K}}{\text{W}} + \frac{1}{10 \frac{\text{W}}{\text{m}^2 \text{K}}}\right) + 298 \text{ K}$$

$$T_h = 400.1 \text{ K} \quad \text{without heat sink}$$

Adding the heat sink reduces the heater temperature by 88°C .

2)

Given

Wire, $D = .001 \text{ m}$, Oil $T_{\infty} = 20^{\circ}\text{C}$, $p = 4000$, $k = 400 \frac{\text{W}}{\text{mK}}$, $c = 500 \frac{\text{J}}{\text{kgK}}$

$R'_c = .01 \Omega/\text{m}$, Oil $h = 500 \text{ W/m}^2\text{K}$

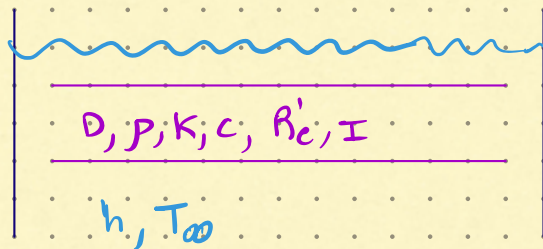
At $t=0$, $I = 100 \text{ A}$ flows

Find

a) SS temp. of wire

b) Find wire temp. as function of time. How long for T_{wire} within 5°C of T_{ss} ?

Schematic



Assumptions

a) SS, 1D heat flow, neglect radiation

Analysis

$$a) \dot{E}_{\text{in}} - \dot{E}_{\text{out}} + \dot{E}_g = \dot{E}_{\text{st}}$$

$$I^2 R'_c L - h(\pi D L)(T - T_{\infty}) = 0$$

$$I^2 R'_c L - h\pi D L T + h\pi D L T_{\infty} = 0$$

$$\Rightarrow T_{\text{ss}} = \frac{I^2 R'_c + h\pi D T_{\infty}}{h\pi D}$$

$$T_{ss} = \frac{(100 \text{ A})^2 (0.01 \text{ m}) + (500 \frac{\text{W}}{\text{m}^2 \text{K}}) \pi (0.001 \text{ m}) (293 \text{ K})}{(500 \frac{\text{W}}{\text{m}^2 \text{K}}) \pi (0.001 \text{ m})}$$

$$T_{ss} = 356.66 \text{ K} = 83.66^\circ \text{C}$$

b)

$$I^2 R_c L - h \pi D L (T - T_\infty) = \rho V_c \frac{dT}{dt}$$

$$V = \pi \left(\frac{D}{2}\right)^2 L$$

At ss, RHS = 0:

$$\Rightarrow \rho V_c \frac{dT}{dt} = h \pi D L (T_{ss} - T_\infty) - h \pi D L (T - T_\infty)$$

T_∞ terms cancel:

$$\Rightarrow \rho V_c \frac{dT}{dt} = h \pi D L (T_{ss} - T)$$

Separating:

$$\frac{dT}{T_{ss} - T} = \frac{h \pi D L}{\rho V_c} dt$$

Integrating:

$$\int \frac{dT}{T_{ss} - T} = \int \frac{h \pi D L}{\rho V_c} dt$$

$$-\ln|T_{ss} - T| = \frac{h \pi D L}{\rho V_c} t + T_0 \xrightarrow{\text{initial temperature}} = T_\infty$$

Rearranging:

$$T(t) = T_{ss} + (T_{\infty} - T_{ss}) e^{-t/\tau}, \text{ where } \tau = \frac{h\pi DL}{\rho V c}$$

Simplifying τ :

$$\tau = \frac{h\pi DL}{\rho V c} = \frac{h\pi DL}{\frac{1}{4}\pi D^2 L \rho c} = \frac{4h}{D\rho c}$$

$$\Rightarrow T(t) = T_{ss} + (T_{\infty} - T_{ss}) e^{-t/\tau}, \quad \tau = \frac{4h}{D\rho c}$$

$$\Rightarrow T(t) = 356.66 - 63.66 e^{-t}, \quad \tau = \frac{4(500)}{4000 \cdot 500 \cdot 0.001} = 1$$

$$351.66 = 356.66 - 63.66 e^{-t}$$

$$\Rightarrow t = 2.54 \text{ seconds} \quad \text{To be within } 5^{\circ}\text{C of ss temp.}$$